Type MS · Describe the products and purposes of mitosis

Answer 1 of 9 · Question 7 part c · 2 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

number				
7 (a)	(i)	• prophase, metaphase, anaphase and telophase;;	minus 1 for each error	2 marks
	(ii)	• chromosomes lined up; • equator shown/aligned at equator; • one member of each bivalent above, the other below;		3 marks
(b)	(i)	A (DNA); B not part of DNA C not part of DNA D found only in RNA not DNA		1 mark
	(ii)	length = 20 mm; $\frac{20}{750}$ = 0.026 mm; = 26.0/26.7 µm;	allow 20 -22 ECF	3 marks
(c)		• two genetically identical cells; • diploid/2n;		2 marks

Total 11 marks

Type MS · Describe the products and purposes of mitosis

Answer 1 of 9 · Question 7 part c · 2 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe the products and purposes of mitosis

Answer 2 of 9 · Question 6 part a.ii · 2 marks

2020 NOV · Paper 01 (regional R) · find part a.ii in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Describe the products and purposes of mitosis

Answer 3 of 9 · Question 6 part a.iii · 2 marks

2020 NOV · Paper 01 (regional R) · find part a.iii in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Describe the products and purposes of mitosis

Answer 4 of 9 · Question 8 part a.ii · 2 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

8 (a) (i)		chromosomes (1)		1
	(ii)	 (1)  (1)		2
(b) (i)		complementary bases pair up 3 DNA double helix unwinds 1 strands separate 2 two DNA strands form 4		1
(c) (i)		thymine/T (1)	R thiamine/thyamine	1
	(ii)	guanine/G/cytosine/C (1)		1
(c)		any four from • allows formation of gametes (1) • haploid (1) • allows variation to occur (1) • which allows species to evolve (1) • allows diploid number to be maintained (1) • at fertilisation (1)		4

Total 10 marks

Type MS · Describe the products and purposes of mitosis

Answer 4 of 9 · Question 8 part a.ii · 2 marks

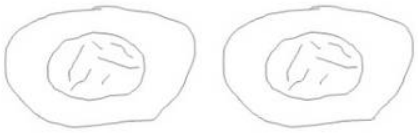
2020 NOV · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

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Type MS · Describe the products and purposes of mitosis

Answer 5 of 9 · Question 2 part b.i · 2 marks

2022 JUN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

2 (a) (i)	156 – 72; 84;	ECF if candidates use incorrect values from the graph with correct answer from for incorrect values = 1 mark	2
(ii)	**C** 9 minutes;		1
(iii)	In the following order only: • oxygen; • diffusion; • aerobic/cellular; • decreases/reduces/goes down;		4
(b) (i)	 must have 6 chromosomes shape not important	One mark for each correct cell drawn	2
(ii)	**B** metaphase; A is incorrect as chromosomes do not align at the equator during anaphase C is incorrect as chromosomes do not align at the equator during prophase D is incorrect as chromosomes do not align at the equator during telophase		1
			Total 10

Type MS · Describe the products and purposes of mitosis

Answer 5 of 9 · Question 2 part b.i · 2 marks

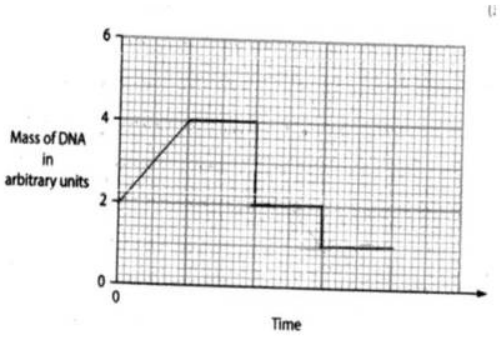
2022 JUN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

uestion number	Answer	Notes	Mark s

Type MS · Describe the products and purposes of mitosis

Answer 6 of 9 · Question 4 part b.ii · 2 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

Number				
4	(a)	(i)	**A; (chromosomes)** B/C/D do not contain DNA	1
		(ii)	RNA/mRNA/tRNA;	1
		(iii)	<table><tr><th>Differences</th><th>DNA</th><th>Named nucleic acid</th></tr><tr><td>1</td><td>double stranded</td><td>single stranded</td></tr><tr><td>2</td><td>contains deoxyribose</td><td>contains ribose</td></tr><tr><td>3</td><td>contains thymine</td><td>contains uracil</td></tr></table>	3
	(b)	(i)	• replication; • formation of a new strand (of DNA); • by complementary base pairing; • double the original mass of DNA;	4
		(ii)	• cell dividing; • cytokinesis;	2
		(iii)	• line starting at two; • increases to four; • ends at one;  	3

Total 14 marks

Type MS · Describe the products and purposes of mitosis

Answer 6 of 9 · Question 4 part b.ii · 2 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe the products and purposes of mitosis

Answer 7 of 9 · Question 2 part b.i · 2 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

2 (a) (i)

Part	Role		
	where sperm are deposited	where eggs are produced	where fertilisation takes place
fallopian tube			✓
ovary		✓	
vagina	✓		

one mark for each correct row

(3)

(ii)

- muscles/uterus wall contract;
- to widen/dilate cervix;

allow description of widening cervix

(2)

(iii)

C (a sperm cell fuses with an egg cell to form a zygote)

The following statements are incorrect because:

A haploid cells are not produced from fertilisation
B an embryo is not formed from fertilisation
D a baby is not formed at fertilisation

(1)

(iv)

luteinising hormone/LH;

(1)

Type MS · Describe the products and purposes of mitosis

Answer 7 of 9 · Question 2 part b.i · 2 marks

2024 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

(b) (i)		One mark for each diagram	(2)
(ii)	mitosis → two cells, each with 23 chromosomes mitosis → four cells, each with 46 chromosomes meiosis → two cells, each with 46 chromosomes meiosis → four cells, each with 23 chromosomes	reject more than one line from each type of cell division	(2)

Total for Question 2 = 11 marks

Type MS · Describe the products and purposes of mitosis

Answer 8 of 9 · Question 2 part b.ii · 2 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b.ii in this mark scheme extract

2 (a) (i)

Part	Role		
	where sperm are deposited	where eggs are produced	where fertilisation takes place
fallopian tube			✓
ovary		✓	
vagina	✓		

one mark for each correct row

(3)

(ii)

- muscles/uterus wall contract;
- to widen/dilate cervix;

allow description of widening cervix

(2)

(iii)

C (a sperm cell fuses with an egg cell to form a zygote)

The following statements are incorrect because:

A haploid cells are not produced from fertilisation
B an embryo is not formed from fertilisation
D a baby is not formed at fertilisation

(1)

(iv)

luteinising hormone/LH;

(1)

Type MS · Describe the products and purposes of mitosis

Answer 8 of 9 · Question 2 part b.ii · 2 marks

2024 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

(b) (i)		One mark for each diagram	(2)
(ii)	mitosis → two cells, each with 23 chromosomes mitosis → four cells, each with 46 chromosomes mitosis → two cells, each with 46 chromosomes meiosis → four cells, each with 23 chromosomes	reject more than one line from each type of cell division	(2)

Total for Question 2 = 11 marks

Type MS · Describe the products and purposes of mitosis

Answer 9 of 9 · Question 2 part b.ii · 1 mark

2025 JUN · Paper 02 (regional R) · find part b.ii in this mark scheme extract

2	(a)	**C (endoplasmic reticulum)** The answer is not A as the cell membrane transports substance into and out of the cell B cytoplasm as this is a base for chemical reactions D as ribosomes provide sites for protein synthesis		1
	(b) (i)	Z Y W X;		1
	(ii)	growth/repair;	Ignore repair of body cells	1
	(c)	Any two from • mitosis produces diploid cells/46 chromosomes, meiosis produces haploid cells/23 chromosomes; • mitosis produces 2 (daughter) cells, meiosis produces 4 cells/gametes; • (daughter) cells from mitosis are genetically identical, (daughter) cells from meiosis show variation;		2
			Total for Question 2 = 5 marks	